TECHNICAL REPORT

Staying Safe on Public Wi-Fi: A Practical Cybersecurity Guide

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Abstract

This report documents the preparation of a public Wi-Fi cybersecurity awareness guide for non-technical users. It explains DNS, HTTPS/TLS, VPNs, firewalls, common public Wi-Fi risks, safe connection practices, and warning signs of suspicious captive portals. The work is research-based and intended for awareness and prevention; no live network penetration testing or laboratory measurements are claimed.

1. Introduction

Public Wi-Fi is commonly available in cafés, hotels, airports, libraries and other shared spaces. It is convenient, but users may not know whether a network is genuine or how to recognise unsafe login prompts. Basic understanding of web security and cautious connection habits can reduce avoidable risks.

This project translates core networking and cybersecurity concepts into plain language for people without a technical background.

2. Project Objective

The objective is to produce a practical, easy-to-understand guide that helps non-technical users connect to public networks more safely and recognise suspicious captive portals.

Specific objectives are to explain DNS, HTTPS/TLS, VPNs and firewalls; describe public Wi-Fi threats; provide a safety checklist; and identify warning signs such as unrelated password requests, unexpected downloads, unfamiliar domains and browser security warnings.

3. Background Research

DNS (Domain Name System) helps resolve domain names to network addresses. If DNS responses or settings are manipulated, a user may be directed to an unintended destination, so checking domain names remains important.

HTTPS uses TLS to protect data exchanged between a browser and a website and helps authenticate the website endpoint. HTTPS does not guarantee that the site itself is honest; phishing websites may also use HTTPS.

A VPN routes device traffic through a VPN server and commonly encrypts traffic between the device and that server. It can provide an additional layer of protection but does not eliminate phishing, malware or unsafe websites.

A firewall filters network traffic according to configured rules and can reduce unwanted connections. It should remain enabled unless a trusted administrator has a specific, justified reason to change it.

Public Wi-Fi risks can include fake hotspots, evil-twin access points, interception of poorly protected traffic, phishing pages, unsafe downloads and unnecessary device sharing. Captive portals are common sign-in pages, but their appearance alone cannot prove legitimacy.

4. Methodology

The work followed a desk-research and content-development approach: (1) identify the intended audience and task requirements; (2) review introductory information from recognised cybersecurity and technology organisations; (3) summarise technical concepts in plain language; (4) convert findings into practical recommendations and a checklist; and (5) organise the guide into clearly labelled sections with references.

This is a documentation and awareness project. No unauthorised network access, penetration testing, packet capture, or live captive-portal testing was performed as part of this report.

5. Work Performed

A guide titled “Staying Safe on Public Wi-Fi: A Practical Cybersecurity Guide” was prepared. It covers DNS, HTTPS/TLS, VPNs, firewalls, public Wi-Fi risks, safe connection steps, suspicious captive-portal warning signs, a quick checklist, a conclusion and reference links.

The practical recommendations include confirming the exact network name with venue staff, checking website domains, keeping devices updated, leaving the firewall enabled, using strong unique passwords and multi-factor authentication, avoiding unknown downloads or certificates, and disconnecting from suspicious networks.

6. Findings

The main finding is that no single security control guarantees safety on public Wi-Fi. Protection is strongest when users combine secure web connections, updated devices, sensible network settings, careful verification of network and website names, and caution around unexpected requests.

HTTPS/TLS protects data in transit to the authenticated website but does not prove that a website is legitimate. A VPN can protect the connection to its server but cannot make a phishing site safe. Firewalls can limit network connections but do not replace user awareness.

7. Evaluation and Testing

The deliverable was reviewed against the assigned topic list: DNS, HTTPS/TLS, VPNs, public Wi-Fi risks, firewalls, safe network connection practices, and suspicious captive portals. The guide also includes a practical checklist and references.

No technical penetration tests, controlled experiments or quantitative measurements were conducted. Therefore, this report does not claim performance results or experimentally verified outcomes. Before submission, the intern should open the final files and confirm that formatting and links display correctly.

8. Limitations

The guide provides general awareness advice and is not a device-specific troubleshooting manual. Menu names and settings vary by operating system and device. The report does not assess a particular Wi-Fi network, VPN provider, firewall configuration or captive portal. Online references and vendor guidance may change over time.

9. Recommendations

Users should verify the exact network name with the venue, use mobile data or another trusted connection for sensitive tasks when practical, inspect website domains, avoid entering unrelated account credentials into Wi-Fi portals, and never bypass browser certificate warnings just to connect.

Keep the operating system, browser and applications updated; use strong unique passwords and multi-factor authentication; keep the firewall enabled; disable unnecessary sharing; and use a reputable VPN where appropriate. If a portal requests a bank PIN, OTP, unknown application or certificate, stop and verify through venue staff using a trusted contact method.

10. Conclusion

This project provides a plain-language cybersecurity resource to help everyday users make safer decisions when using public Wi-Fi. It explains the roles and limits of DNS, HTTPS/TLS, VPNs and firewalls and gives actionable advice for recognising suspicious captive portals. Continued awareness and careful verification are important because technical controls reduce risk but cannot eliminate every threat.

References

Cloudflare, “What is DNS?” — https://www.cloudflare.com/learning/dns/what-is-dns/

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Cybersecurity and Infrastructure Security Agency (CISA) — https://www.cisa.gov/